22.6.3 Shale gas

Shale refers to the **soft finely stratified sedimentary rock** that is formed from the compaction of small old rocks containing mud and minerals – such as quartz and calcite, trapped beneath earth's surface. These rocks contain fossil fuels like oil and gas in their pores.

The fuel is extracted by a technique called **hydraulic fracturing** (drilling or well boring of sedimentary rocks layers to reach productive reservoir layers).

Environmental concerns of shale gas

(i) Shale drilling could affect groundwater reserves, which can contaminate the drinking water resources and also affect the fertility of the soil.

(ii) Million gallons of water is needed to break and release the shale gas, which inturn can affect the water table.

More to Know

India has identified six basins as areas for shale gas exploration: Cambay (Gujarat), Assam-Arakan (North East), Gondwana (Central India), Krishna Godavari onshore (East Coast), Cauvery onshore and Indo-Gangetic basins.

22.6.4 Wind Energy

The **kinetic energy** possessed by the wind is due to its high speed, that can be **converted into mechanical power by wind turbines**. The rotatory motion of wind mill produces wind energy. It can be used for generating electricity, run water pumps, flour mills, draw water from wells etc.,

DO YOU KNOW?

- The world's largest and tallest wind turbine is situated in Hawaii.
- One wind turbine can produce electricity for 300 homes.

Windmill

Windmill is a machine that converts the energy of wind into rotational energy by broad blade attached to the rotating axis. When the blowing air strikes the blades of the windmill, it exerts force and causes the blades to rotate. The rotational movement of the blades operate the generator and the electricity is produced. The energy output from each windmill is coupled together to get electricity on a commercial scale.



Figure 22.2 Windmill

Advantages of Wind energy

(i) Wind energy is free, eco-friendly, renewable source of energy.

(ii) It does not cause pollution.

(iii) Expenses on periodic maintenance is low when compared to the other power sources.

Activity 1

Collect information regarding the

(i) Tehri Dam project

(ii) Sardar Sarovar Dam project

22.6.5 Water Energy

Earth's surface is covered with nearly 71 % of water. **Harnessing the energy from the flowing water** can be used to produce electricity. The technique to harness the **water energy** is called **Hydropower**.

The electrical energy is derived from water flow, water falling from a height. Hilly areas are suitable for this purpose where there is continous flow of water in large amounts falling from high slopes. It does not cause environmental pollution or waste generation.

10th_Science Unit-22.indd 321 12/3/2022 3:13:04 PM